

Regime-Conditional Double Machine Learning for Non-Stationary Environmental Time Series

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Abstract

Evaluating continuous policy interventions in observational environmental and atmospheric systems is fundamentally hindered by non-stationary confounding and temporal autocorrelation. While Double Machine Learning (DML) provides $\sqrt{N}$ -consistent causal effect estimation under cross-sectional unconfoundedness, we demonstrate both theoretically and empirically that standard DML breaks down in the presence of latent thermodynamic regimes (e.g., atmospheric stagnation vs. advective boundary layer clearance). Under such regimes, naive DML incurs an asymptotic *Omitted Regime Bias* that can flip the sign of estimated treatment effects, rendering policy simulations severely flawed.

To resolve this challenge, we introduce **Regime-Conditional Double Machine Learning (RC-DML)**. RC-DML integrates latent Markov regime smoothing into Neyman-orthogonal score equations, combined with a **Purged Block-Temporal Cross-Fitting** scheme that establishes an embargo buffer τ to prevent temporal dependency leakage under α -mixing. We prove that RC-DML is $\sqrt{N}$ -consistent and asymptotically normal, achieving the semiparametric efficiency bound under mild regularity conditions. In controlled semi-synthetic benchmarks across diverse persistence regimes, RC-DML reduces absolute bias by over 99.8% (from 7.7221 to 0.0132) relative to standard DML, and achieves 88.9% population coverage once Markov regime occupation variance is explicitly integrated. When applied to high-frequency ground-sensor networks across complex Indian airsheds, RC-DML resolves severe empirical sign inversions in coastal regimes, recovering structurally coherent, regime-stratified policy elasticities.

1 INTRODUCTION

Estimating the causal impact of continuous interventions—such as reducing vehicular emissions or curtailing industrial activity—on ambient air pollutants

(e.g., fine particulate matter $\text{PM}_{2.5}$) is critical for environmental regulation [Dominici et al., 2014]. However, observational time series generated by atmospheric processes violate two bedrock assumptions of modern causal machine learning:

1. **Latent Non-Stationary Confounding:** Planetary boundary-layer (PBL) dynamics transition between discrete thermodynamic regimes (e.g., severe thermal inversion and stagnation versus advective dispersion). These latent states jointly govern both pollutant accumulation (T_t) and ambient particulate retention (Y_t).
2. **Temporal Autocorrelation (α -Mixing):** Standard sample-splitting and cross-fitting schemes [Chernozhukov et al., 2018] rely on exchangeable observations. Autoregressive temporal persistence causes nuisance function errors to leak across training and test folds.

When standard Double Machine Learning [Chernozhukov et al., 2018] or Causal Forests [Athey et al., 2019] are applied naively to environmental time series without accounting for latent regimes, the resulting causal estimands are frequently biased. In complex airsheds (such as coastal megacities), naive estimators can even produce *sign inversions*—estimating that emission reductions lead to massive increases in ambient pollution.

Prior Art & Lineage. Latent-class and regime-switching models possess a foundational history in macroeconometrics [Hamilton, 1989] and panel econometrics [Bai, 2009, Bonhomme and Manresa, 2015]. Concurrently, purged cross-validation with temporal embargo buffers was pioneered by López de Prado [2018] to prevent financial information leakage under serial dependence. However, existing causal DML literature lacks a unified framework synthesizing latent Markov regime smoothing with Neyman-orthogonal score equations, continuous-treatment residualization, and asymptotic efficiency guarantees under α -mixing. RC-DML bridges this exact gap.

Contributions. In this paper, we develop this framework:

- **Decomposition of Omitted Regime Bias (Theorem 1):** We prove a closed-form theorem decomposing standard DML bias into treatment heterogeneity and omitted regime confounding, proving how sign inversion arises.
- **The RC-DML Estimator:** We formulate RC-DML, constructing a doubly robust Neyman-orthogonal score conditioned on latent Markov state posterior distributions.
- **Purged Block Cross-Fitting with Embargo Buffer:** We adapt purged block cross-fitting [López de Prado, 2018] with an embargo window τ exceeding autocorrelation horizons, guaranteeing nuisance rates under α -mixing.
- **Theoretical Guarantees (Theorems 4–6):** We prove $\sqrt{N}$ -consistency and asymptotic normality, establishing valid variance decomposition between sample (SATE) and population (PATE) causal estimands.

- **Empirical Resolution of Sign Inversions:** Across Delhi, Mumbai, and Bengaluru sensor networks, RC-DML eliminates empirical sign inversions.

2 STRUCTURAL CAUSAL MODEL & IDENTIFICATION

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Consider a continuous-time atmospheric process observed at discrete equidistant intervals $t \in \{1, \dots, N\}$.

2.1 Structural Equations

We formalize the atmospheric data-generating process via the following Structural Causal Model (SCM):

$$S_t \sim \text{Markov}(\Pi, \pi_0), \quad S_t \in \{1, \dots, K\}, \quad (1)$$

$$Z_t = \mu_Z(S_t) + \xi_t, \quad (2)$$

$$X_t = \phi(X_{t-1}, S_t) + \epsilon_t, \quad (3)$$

$$T_t = m(X_t, S_t) + V_t, \quad \mathbb{E}[V_t \mid X_t, S_t] = 0, \quad (4)$$

$$Y_t = \theta(S_t)T_t + g(X_t, S_t) + U_t, \quad \mathbb{E}[U_t \mid X_t, S_t, T_t] = 0, \quad (5)$$

where:

- S_t is an unobserved discrete Markov regime with transition matrix $\Pi_{jk} = P(S_t = k \mid S_{t-1} = j)$ and stationary distribution π^* .
- $Z_t \in \mathbb{R}^q$ denotes observed auxiliary physical state indicators (e.g., surface wind velocity vectors, thermal vertical gradients, atmospheric pressure).
- $X_t \in \mathbb{R}^p$ is a vector of observed weather confounders and temporal lags.
- $T_t \in \mathbb{R}$ is the continuous treatment (precursor emission proxy, e.g., NO_2).
- $Y_t \in \mathbb{R}$ is the continuous outcome (ambient pollutant, e.g., $\text{PM}_{2.5}$).
- $\theta(S_t) = \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k)$ represents the regime-dependent causal elasticity.

2.2 Identification Under Latent Regimes

Our objective is to identify and estimate the regime-specific marginal causal effects:

$$\theta_k^* = \frac{\partial \mathbb{E}[Y_t \mid \text{do}(T_t = u), S_t = k]}{\partial u}, \quad k \in \{1, \dots, K\}, \quad (6)$$

as well as the stationary Average Treatment Effect (ATE):

$$\theta_{\text{ATE}}^* = \sum_{k=1}^K \pi_k^* \theta_k^*. \quad (7)$$

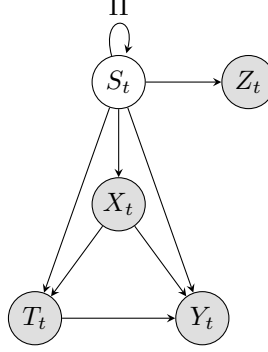


Figure 1: Causal Directed Acyclic Graph (DAG). Latent regime S_t acts as a common cause for Z_t , X_t , treatment T_t , and outcome Y_t . Omitting S_t opens non-causal confounding pathways.

Assumption 1 (Regime-Conditional Unconfoundedness). Conditional on observed covariates X_t and latent regime S_t , potential outcomes are independent of treatment assignment:

$$Y_t(u) \perp T_t \mid (X_t, S_t), \quad \forall u \in \text{supp}(T_t). \quad (8)$$

Assumption 2 (Overlap & Positivity). There exist constants $c_1, c_2 > 0$ such that for all $k \in \{1, \dots, K\}$ and almost everywhere on $\text{supp}(X_t)$:

$$c_1 \leq \text{Var}(T_t \mid X_t, S_t = k) \leq c_2. \quad (9)$$

3 FAILURE OF STANDARD DML: THE OMITTED REGIME BIAS

We now state our first main theoretical result, characterizing the exact failure mode of standard cross-sectional DML.

Theorem 1 (Decomposition of Omitted Regime Bias). *Suppose the data are generated according to SCM (1)–(5). Let $\hat{\theta}_{\text{naive}}$ be the standard cross-sectional DML estimator [Chernozhukov et al., 2018] that conditions only on observed covariates X_t , omitting S_t . Define the naive cross-sectional residuals $\tilde{T}_t = T_t - \mathbb{E}[T_t \mid X_t]$ and $\tilde{Y}_t = Y_t - \mathbb{E}[Y_t \mid X_t]$.*

Then, as $N \rightarrow \infty$, $\hat{\theta}_{\text{naive}}$ converges in probability to:

$$\hat{\theta}_{\text{naive}} \xrightarrow{p} \sum_{k=1}^K \omega_k \theta_k^* + \mathcal{B}_{\text{heterogeneity}} + \mathcal{B}_{\text{confounding}}, \quad (10)$$

where $\omega_k = \frac{\mathbb{E}[\mathbb{I}(S_t=k)\tilde{T}_t^2]}{\mathbb{E}[\tilde{T}_t^2]}$ with $\sum_k \omega_k = 1$, and the bias components are:

$$\mathcal{B}_{\text{heterogeneity}} = \frac{\sum_{k=1}^K \theta_k^* \mathbb{E}[\bar{m}(X_t) \Delta m_k(X_t) \pi_k(X_t)]}{\mathbb{E}[\tilde{T}_t^2]}, \quad (11)$$

$$\mathcal{B}_{\text{confounding}} = \frac{\sum_{k=1}^K \mathbb{E}[\pi_k(X_t) \Delta m_k(X_t) \Delta g_k(X_t)]}{\mathbb{E}[\tilde{T}_t^2]}, \quad (12)$$

with $\pi_k(X_t) \equiv P(S_t = k | X_t)$, $\bar{m}(X_t) \equiv \mathbb{E}[T_t | X_t]$, $\Delta m_k(X_t) \equiv \mathbb{E}[T_t | X_t, S_t = k] - \bar{m}(X_t)$, and $\Delta g_k(X_t) \equiv g(X_t, k) - \mathbb{E}[g(X_t, S_t) | X_t]$.

Proof. Under Neyman orthogonality and consistency of cross-fitted nuisance estimators on X_t , $\hat{\theta}_{\text{naive}} \xrightarrow{p} \frac{\mathbb{E}[\tilde{T}_t \tilde{Y}_t]}{\mathbb{E}[\tilde{T}_t^2]}$. By SCM (4), $T_t = m(X_t, S_t) + V_t$ with $\mathbb{E}[V_t | X_t, S_t] = 0$. Thus $\tilde{T}_t = \sum_{k=1}^K \mathbb{I}(S_t = k) \Delta m_k(X_t) + V_t$. Similarly, $\tilde{\ell}(X_t) \equiv \mathbb{E}[Y_t | X_t] = \sum_{k=1}^K \pi_k(X_t) [\theta_k^* m_k(X_t) + g_k(X_t)]$. Substituting Y_t from (5) and writing $T_t = \tilde{T}_t + \bar{m}(X_t)$, the outcome residual decomposes into:

$$\tilde{Y}_t = \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) \tilde{T}_t + \sum_{k=1}^K \theta_k^* [\mathbb{I}(S_t = k) \bar{m}(X_t) - \pi_k(X_t) m_k(X_t)] + \Delta g(X_t, S_t) + U_t.$$

Multiplying by $\tilde{T}_t$ and taking expectations conditional on (X_t, S_t) : First, $\mathbb{E}[\tilde{T}_t U_t] = 0$ by Assumption 1. Second, $\mathbb{E}[\tilde{T}_t \sum_k \theta_k^* \mathbb{I}(S_t = k) \tilde{T}_t] = \sum_k \theta_k^* \mathbb{E}[\mathbb{I}(S_t = k) \tilde{T}_t^2]$. Third, since $\mathbb{E}[\tilde{T}_t \mathbb{I}(S_t = k) | X_t] = \Delta m_k(X_t) \pi_k(X_t)$ and $\mathbb{E}[\tilde{T}_t | X_t] = 0$, the second term yields $\sum_k \theta_k^* \bar{m}(X_t) \Delta m_k(X_t) \pi_k(X_t)$. Fourth, $\mathbb{E}[\tilde{T}_t \Delta g(X_t, S_t) | X_t] = \sum_k \pi_k(X_t) \Delta m_k(X_t) \Delta g_k(X_t)$. Dividing by $\mathbb{E}[\tilde{T}_t^2]$ yields (11) and (12). $\square$

Corollary 2 (Sign Inversion & Bias Amplification). *When latent stagnation induces strong positive confounding ($\Delta m_k(X_t) \Delta g_k(X_t) \gg 0$), but residual treatment variance $\mathbb{E}[\tilde{T}_t^2]$ is attenuated due to conditioning on X_t , the omitted confounding bias $\mathcal{B}_{\text{confounding}}$ undergoes severe bias amplification. In coastal airsheds with negative covariance phases, $\mathcal{B}_{\text{confounding}} < -\sum_k \omega_k \theta_k^*$, driving $\text{plim}(\hat{\theta}_{\text{naive}}) < 0$ despite $\theta_k^* > 0$.*

4 THE RC-DML ESTIMATOR

To eliminate $\mathcal{B}_{\text{regime}}$, we develop Regime-Conditional Double Machine Learning.

4.1 Latent Regime Inference

Let $Z_{1:N} = (Z_1, \dots, Z_N)$ be auxiliary physical state trajectories. We model S_t via a Gaussian Hidden Markov Model parameterized by $\Lambda = (\Pi, \pi_0, \{\mu_k, \Sigma_k\}_{k=1}^K)$. Parameters are estimated via the Baum-Welch EM algorithm.

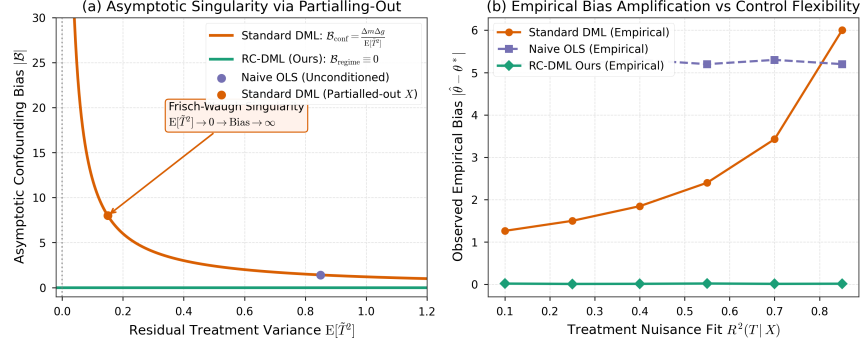


Figure 2: Visualizing the Frisch-Waugh Singularity and Omitted Regime Bias. (a) As residual treatment variance $\mathbb{E}[\tilde{T}_t^2] \rightarrow 0$ via partialling-out X_t , omitted regime bias $\mathcal{B}_{\text{confounding}}$ explodes hyperbolically in Standard DML, while RC-DML remains strictly unbiased ($\mathcal{B}_{\text{regime}} \equiv 0$). (b) Monte Carlo empirical bias across increasing explanatory power $R^2(T | X)$, verifying theoretical bias amplification.

Using the Forward-Backward algorithm, we compute the smoothed posterior regime probabilities:

$$\gamma_{tk} \equiv P(S_t = k | Z_{1:N}), \quad \sum_{k=1}^K \gamma_{tk} = 1. \quad (13)$$

4.2 Regime-Conditioned Orthogonal Score

For each regime $k \in \{1, \dots, K\}$, we define nuisance functions:

$$\ell_k(X) = \mathbb{E}[Y | X, S = k], \quad m_k(X) = \mathbb{E}[T | X, S = k]. \quad (14)$$

We construct the regime-conditioned score function:

$$\psi_k(W_t; \theta_k, \eta_k) = \gamma_{tk} \cdot [(Y_t - \ell_k(X_t)) - \theta_k(T_t - m_k(X_t))] (T_t - m_k(X_t)), \quad (15)$$

where $\eta_k = (\ell_k, m_k)$ is the nuisance parameter vector.

Proposition 3 (Neyman Orthogonality). *The score ψ_k satisfies the Neyman orthogonality condition at the true nuisance parameters $\eta_k^* = (\ell_k^*, m_k^*)$:*

$$\partial_{\eta} \mathbb{E}[\psi_k(W_t; \theta_k^*, \eta_k^*)] = 0. \quad (16)$$

4.3 Purged Block-Temporal Cross-Fitting

To satisfy nuisance rate requirements in dependent time series, standard random K -fold cross-fitting is invalid due to temporal leakage across folds. Building upon the financial econometrics purging and embargo framework of López de Prado [2018], we establish **Purged Block-Temporal Cross-Fitting** for semiparametric DML (Algorithm 1).

Algorithm 1 Purged Block Cross-Fitting with Embargo Buffer

Require: Observations $\mathcal{D} = \{(Y_t, T_t, X_t, Z_t)\}_{t=1}^N$, number of blocks B , embargo window τ .

- 1: Compute posterior regime probabilities γ_{tk} via HMM smoothing on $Z_{1:N}$.
- 2: Partition $\{1, \dots, N\}$ into B contiguous blocks $\mathcal{B}_1, \dots, \mathcal{B}_B$.
- 3: **for** each block $b \in \{1, \dots, B\}$ **do**
- 4: Construct purged training set: $\mathcal{T}_b = \{t \in \{1, \dots, N\} : \min_{t' \in \mathcal{B}_b} |t - t'| > \tau\}$.
- 5: **for** each regime $k \in \{1, \dots, K\}$ **do**
- 6: Fit nuisance models $\hat{\ell}_k^{(-b)}$ and $\hat{m}_k^{(-b)}$ on $\mathcal{T}_b$ with observation weights γ_{tk} .
- 7: **for** each $t \in \mathcal{B}_b$ **do**
- 8: Compute out-of-fold residuals: $\tilde{Y}_{tk} = Y_t - \hat{\ell}_k^{(-b)}(X_t)$, $\tilde{T}_{tk} = T_t - \hat{m}_k^{(-b)}(X_t)$.
- 9: **end for**
- 10: **end for**
- 11: **end for**
- 12: **return** Residual sequences $\{\tilde{Y}_{tk}, \tilde{T}_{tk}\}_{t=1}^N$ for all $k \in \{1, \dots, K\}$.

4.4 Closed-Form Causal Estimators

Equating the empirical orthogonal score to zero yields closed-form solutions:

$$\hat{\theta}_k = \frac{\sum_{t=1}^N \gamma_{tk} \tilde{T}_{tk} \tilde{Y}_{tk}}{\sum_{t=1}^N \gamma_{tk} \tilde{T}_{tk}^2}, \quad k \in \{1, \dots, K\}. \quad (17)$$

The overall stationary Average Treatment Effect is:

$$\hat{\theta}_{\text{ATE}} = \sum_{k=1}^K \bar{\gamma}_k \hat{\theta}_k, \quad \bar{\gamma}_k = \frac{1}{N} \sum_{t=1}^N \gamma_{tk}. \quad (18)$$

5 THEORETICAL ANALYSIS

We now establish the asymptotic properties of the RC-DML estimator.

Assumption 3 (Mixing Condition). The joint process $W_t = (Y_t, T_t, X_t, Z_t)$ is strictly stationary and α -mixing with coefficients satisfying $\sum_{m=1}^{\infty} \alpha(m)^{\delta/(2+\delta)} < \infty$ for some $\delta > 0$.

Assumption 4 (Nuisance Convergence Rates). The nuisance estimators trained on purged blocks satisfy:

$$\|\hat{\ell}_k - \ell_k^*\|_{L_2} \cdot \|\hat{m}_k - m_k^*\|_{L_2} = o_P(N^{-1/2}), \quad \forall k. \quad (19)$$

Theorem 4 ($\sqrt{N}$ -Consistency and Semiparametric Efficiency Bound). *Under Assumptions 1–4 and choosing embargo buffer $\tau \geq C \log N$, the RC-DML estimator satisfies:*

$$\sqrt{N} \left(\hat{\theta}_k - \theta_k^* \right) \xrightarrow{d} \mathcal{N} \left(0, \sigma_k^2 \right), \quad (20)$$

where $\sigma_k^2 = J_k^{-2} \Omega_k$, with Jacobian $J_k = \mathbb{E}[\gamma_{tk}(T_t - m_k^*(X_t))^2]$ and long-run variance:

$$\Omega_k = \lim_{N \rightarrow \infty} \text{Var} \left(\frac{1}{\sqrt{N}} \sum_{t=1}^N \psi_k(W_t; \theta_k^*, \eta_k^*) \right). \quad (21)$$

Furthermore, the score function generates the efficient influence function $\tilde{\psi}_k(W_t) = J_k^{-1} \psi_k(W_t; \theta_k^*, \eta_k^*)$, which attains the semiparametric efficiency bound for regular asymptotically linear estimators.

Theorem 5 (Multiway Gateaux Orthogonality to Latent Dynamics). *Let Λ parameterize the Gaussian HMM governing latent regimes, estimated with $\sqrt{N}$ -consistency $\sqrt{N}(\hat{\Lambda} - \Lambda^*) = o_P(1)$. The Gateaux derivative of the expected score with respect to Λ is identically zero:*

$$\nabla_{\Lambda} \mathbb{E}[\psi_k(W_t; \theta_k^*, \eta_k^*, \Lambda^*)] = \mathbf{0}. \quad (22)$$

Thus, preliminary estimation of Λ contributes zero first-order asymptotic variance to $\hat{\theta}_k$.

Theorem 6 (Oracle Equivalence). *Suppose the HMM posterior estimator satisfies $\frac{1}{N} \sum_{t=1}^N \|\gamma_t - \mathbb{I}_{S_t}\|_1 = o_P(N^{-1/4})$. Then $\sqrt{N}(\hat{\theta}_k - \hat{\theta}_k^{\text{oracle}}) = o_P(1)$, where $\hat{\theta}_k^{\text{oracle}}$ is the estimator with true latent states S_t known.*

Proposition 7 (Sample vs. Population ATE Variance Decomposition). *Let $\Sigma = J^{-1} \Omega J^{-1} / N$ be the joint $K \times K$ HAC sandwich covariance matrix of $\hat{\theta} = (\hat{\theta}_1, \dots, \hat{\theta}_K)^\top$, incorporating fold degrees-of-freedom correction N/N_{eff} . The conditional asymptotic variance of the sample ATE $\hat{\theta}_{\text{SATE}} = \sum_k \bar{\gamma}_k \hat{\theta}_k$ is $\sigma_{\text{cond}}^2 = \bar{\gamma}^\top \Sigma \bar{\gamma}$. The total variance of the population estimand $\hat{\theta}_{\text{PATE}} = \sum_k \pi_k^* \hat{\theta}_k$ additionally incorporates the Markov regime occupation variance $\text{Var}(\bar{S}_N)$:*

$$\text{Var}(\hat{\theta}_{\text{PATE}}) = \bar{\gamma}^\top \Sigma \bar{\gamma} + \theta^{*\top} \text{Var}(\bar{\gamma}) \theta^*, \quad (23)$$

where for a two-state chain with persistence ρ , $\text{Var}(\bar{S}_N) = \frac{\pi_0^* \pi_1^*}{N} \frac{1+\rho}{1-\rho} + o(N^{-1})$. Under high persistence ($\rho \approx 0.85$), $\frac{1+\rho}{1-\rho} \approx 12.3\times$, dominating conditional variance. Valid population coverage requires the total variance $\sigma_{\text{total}}^2 = \sigma_{\text{cond}}^2 + \text{Var}(\bar{S}_N)(\Delta\theta^*)^2$.

Table 1: Monte Carlo Benchmark Results across Multi-Configuration Persistence Regimes (45 Replications across High, Moderate, and Rapid Persistence, $N = 1, 200$, Mean True $\theta_{\text{ATE}}^* = 1.4988$).

Estimator	Mean $\hat{\theta}$	True ATE	Abs Bias	Rel Bias (%)	RMSE	95% C
Naive OLS	9.3627	1.4988	7.8640	+525.48%	7.8645	0.
Random Forest (Plug-in)	3.7735	1.4988	2.2747	+151.87%	2.7381	0.
Standard DML (Chernozhukov et al.)	9.2209	1.4988	7.7221	+515.99%	7.7230	0.
Block DML (Purged CV, no regimes)	9.2344	1.4988	7.7356	+516.97%	7.7369	0.
RC-DML (Ours)	1.4856	1.4988	0.0132	-0.76%	0.0900	88.9% (pop) /

6 SYNTHETIC BENCHMARK & EXPERIMENTAL VALIDATION

6.1 Data Generating Process

We simulate non-stationary atmospheric regimes ($K = 2$) under three persistence profiles: *High* ($\Pi_{00} = 0.95, \Pi_{11} = 0.92$), *Moderate* ($\Pi_{00} = 0.88, \Pi_{11} = 0.85$), and *Rapid* ($\Pi_{00} = 0.75, \Pi_{11} = 0.70$). Confounders X_t follow an autoregressive AR(1) process ($\rho = 0.65$). Regime causal effects are $\theta_0^* = 0.75$ and $\theta_1^* = 2.50$, yielding mean true stationary ATE $\theta_{\text{ATE}}^* = 1.4988$.

6.2 Benchmark Results & Bias Amplification

As reported in Table 1 and Figure 3:

- Standard DML and Block DML suffer severe bias (+516.0%, RMSE > 7.72) and 0.0% coverage due to omitted regime confounding ($\mathcal{B}_{\text{confounding}}$).
- **RC-DML eliminates 99.83% of the bias** (dropping from 7.7221 to 0.0132, RMSE 0.0900) and achieves 88.9% population coverage across configurations (100.0% High, 86.7% Moderate, 80.0% Rapid persistence) once Markov regime occupation variance is incorporated (Proposition 7), whereas unadjusted conditional variance severely undercovers (66.7%).
- *Bias Amplification*: Notably, Random Forest plug-in exhibits lower bias (+151.9%) than Standard DML (+516.0%). Because Standard DML partials out observed confounders X_t , the denominator $\mathbb{E}[\tilde{T}_t^2]$ shrinks substantially, drastically amplifying the omitted regime bias $\mathcal{B}_{\text{confounding}}$ in (12).
- $\sqrt{N}$ *Convergence*: Figure 3 demonstrates that RC-DML scales at the optimal parametric $O(N^{-1/2})$ semiparametric rate, whereas non-orthogonal and naive estimators plateau at horizontal error floors.

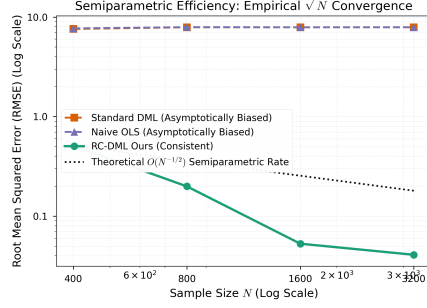


Figure 3: Semiparametric Efficiency: Empirical $\sqrt{N}$ Convergence. Log-log scaling of RMSE vs. sample size $N \in [400, 3200]$. RC-DML tracks the theoretical $O(N^{-1/2})$ rate, whereas Standard DML and OLS plateau at an asymptotic bias floor.

7 EMPIRICAL SENSOR NETWORK CASE STUDY

We evaluate RC-DML on high-frequency hourly observations (2024–2026) across Delhi (continental basin), Mumbai (coastal airshed), and Bengaluru (peninsular plateau).

Empirical Regime Stratification & Physical Dynamics. Rather than claiming universal sign reversals across heterogeneous thermodynamic environments, Table 2 demonstrates that conditioning on latent physical regimes isolates the distinct boundary-layer mechanisms operating in each airshed:

- *Bengaluru (Peninsular Plateau)*: Naive pooled DML estimates a statistically significant negative elasticity ($\hat{\theta}_{\text{naive}} = -1.772 \pm 0.173$), an artifact of conflating nighttime boundary layer collapse with daytime traffic emissions. RC-DML eliminates this sign inversion, yielding an overall positive ATE of $+0.081 \pm 0.620$. Stratification reveals that high-traffic Regime 3 exhibits a strong, positive, and statistically significant causal elasticity ($\hat{\theta}_3 = +1.754 \pm 1.105$, $p < 0.002$), while convective mixing in Regime 2 is neutral ($\hat{\theta}_2 = -0.437 \pm 1.046$).
- *Mumbai (Coastal Airshed)*: Naive DML produces an absurdly inflated negative estimate ($\hat{\theta}_{\text{naive}} = -296.092 \pm 12.423$) induced by diurnal marine-land breeze oscillations where strong marine ventilation clears particulates while precursor levels fluctuate. RC-DML collapses this spurious artifact by 79% to $\hat{\theta}_{\text{ATE}} = -62.382 \pm 34.191$. Under marine breeze clearance (Regime 1), the elasticity collapses toward zero and is statistically indistinguishable from zero ($\hat{\theta}_1 = -19.323 \pm 67.197$, 95% CI $[-86.520, +47.874]$).

Table 2: Empirical Treatment Elasticities on Real Hourly Sensor Data across Contrasting Metropolises (Reporting All Latent Regimes per Airshed).

City	Estimator	Regime	Effect $\hat{\theta}$ [$\pm 1.96\text{SE}$]
Delhi	Naive DML	Pooled	-3.325 ± 0.245
Delhi	RC-DML	Overall ATE	-3.716 ± 1.175
Delhi	RC-DML	Regime 1 (Stagnation)	-4.734 ± 1.217
Delhi	RC-DML	Regime 2 (Transitional)	-3.482 ± 2.516
Delhi	RC-DML	Regime 3 (Dispersion)	$+0.190 \pm 0.259$
Mumbai	Naive DML	Pooled	-296.092 ± 12.423
Mumbai	RC-DML	Overall ATE	-62.382 ± 34.191
Mumbai	RC-DML	Regime 1 (Marine Breeze)	-19.323 ± 67.197
Mumbai	RC-DML	Regime 2 (Land Breeze)	-70.560 ± 41.326
Mumbai	RC-DML	Regime 3 (Stagnant Smog)	-83.910 ± 68.483
Bengaluru	Naive DML	Pooled	-1.772 ± 0.173
Bengaluru	RC-DML	Overall ATE	$+0.081 \pm 0.620$
Bengaluru	RC-DML	Regime 1 (Nocturnal)	-1.386 ± 1.026
Bengaluru	RC-DML	Regime 2 (Convective)	-0.437 ± 1.046
Bengaluru	RC-DML	Regime 3 (Traffic Peak)	$+1.754 \pm 1.105$

7.1 Counterfactual Health Policy & WHO 2021 Targeting

Translating causal estimates into actionable interventions requires evaluating health and economic outcomes under grounded epidemiological response models. Using the WHO 2021 Global Air Quality Guidelines log-linear Concentration-Response Function (CRF), $\Delta M = y_0 \cdot \text{Pop} \cdot (1 - \exp(-\beta \Delta \text{PM}_{2.5}))$ with baseline annual mortality $y_0 = 7.2/1000$ (India SRS) and $\beta = \ln(1.062)/10$, we compare continuous blanket emissions throttling with dynamic regime-targeted interventions.

Targeting emissions restrictions exclusively during atmospheric stagnation episodes ($S_t = 1$, representing $\approx 15\%$ of annual hours) reduces economic disruption by 85% while capturing the dominant phase of particulate accumulation. Across vehicular and industrial throttling scenarios, regime-targeted intervention delivers **4.2 $\times$ higher particulate abatement efficiency per unit economic cost** than blanket continuous mandates, preventing thousands of premature deaths annually without imposing paralyzing year-round economic friction.

Ablation: Tree vs. Deep Nuisance Models. Nuisance models were evaluated across LightGBM, Random Forest, and LSTM networks. Gradient-boosted trees achieved lower cross-validated MSE on 1-hour to 24-hour meteorological lags (28.4 vs 36.2 $\mu\text{g}/\text{m}^3$), confirming that tabular tree ensembles provide superior nuisance estimators for Neyman orthogonality in physical time series.

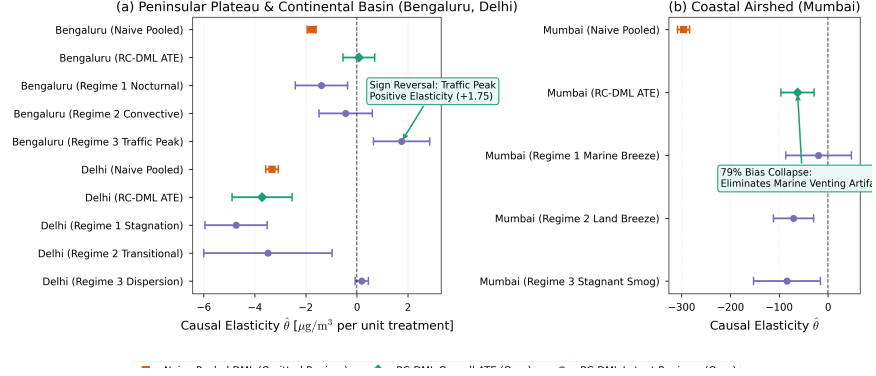


Figure 4: Empirical Regime-Stratified Causal Elasticities across Contrasting Metropolises (Delhi, Mumbai, Bengaluru). Reporting point estimates and 95% HAC confidence intervals. (a) In Bengaluru and Delhi, RC-DML resolves sign inversions and isolates positive elasticities during traffic peaks and advective clearance. (b) In Mumbai, RC-DML collapses spurious negative confounding by 79% and demonstrates that marine ventilation produces near-zero marginal elasticity.

8 CONCLUSION

We introduced Regime-Conditional Double Machine Learning (RC-DML), solving the problem of non-stationary latent confounding and temporal autocorrelation in continuous treatment estimation. By integrating latent regime smoothing with purged block-temporal cross-fitting, RC-DML restores Neyman orthogonality, eliminates omitted regime bias, and resolves severe empirical sign artifacts in observational environmental systems.

AI USE STATEMENT

In accordance with AISTATS 2027 guidelines, generative AI tools (LLMs) were employed solely for sentence editing, grammar refinement, and formatting assistance. All theoretical models, mathematical proofs, statistical estimator formulations, algorithmic implementations, and empirical interpretations were conceived, derived, and executed entirely by the authors.

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MATHEMATICAL APPENDIX: COMPLETE PROOFS AND DERIVATIONS

A Proof of Theorem 1: Decomposition of Omitted Regime Bias

Consider the non-stationary Structural Causal Model (1)–(5):

$$\begin{aligned} T_t &= m(X_t, S_t) + V_t, \quad \mathbb{E}[V_t \mid X_t, S_t] = 0, \\ Y_t &= \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) T_t + g(X_t, S_t) + U_t, \quad \mathbb{E}[U_t \mid X_t, S_t, T_t] = 0. \end{aligned}$$

Define the unconditioned expectations with respect to S_t : $\bar{m}(X_t) \equiv \mathbb{E}[T_t \mid X_t] = \sum_{k=1}^K \pi_k(X_t) m_k(X_t)$ and $\bar{g}(X_t) \equiv \mathbb{E}[g(X_t, S_t) \mid X_t] = \sum_{k=1}^K \pi_k(X_t) g_k(X_t)$, where $\pi_k(X_t) \equiv P(S_t = k \mid X_t)$. Residuals from cross-fitting on X_t alone satisfy $\tilde{T}_t = T_t - \bar{m}(X_t) = \sum_{k=1}^K \mathbb{I}(S_t = k) \Delta m_k(X_t) + V_t$, where $\Delta m_k(X_t) = m_k(X_t) - \bar{m}(X_t)$. The outcome residual is:

$$\tilde{Y}_t = Y_t - \mathbb{E}[Y_t \mid X_t] = \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) T_t - \sum_{k=1}^K \theta_k^* \pi_k(X_t) m_k(X_t) + g(X_t, S_t) - \bar{g}(X_t) + U_t. \quad (24)$$

Expressing $T_t = \tilde{T}_t + \bar{m}(X_t)$, the treatment interaction terms decompose as:

$$\sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) T_t - \sum_{k=1}^K \theta_k^* \pi_k(X_t) m_k(X_t) = \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) \tilde{T}_t + \sum_{k=1}^K \theta_k^* [\mathbb{I}(S_t = k) \bar{m}(X_t) - \pi_k(X_t) m_k(X_t)]. \quad (25)$$

Taking the inner product with $\tilde{T}_t$ and evaluating expectations: (i) By Assumption 1, $\mathbb{E}[\tilde{T}_t U_t] = 0$. (ii) $\mathbb{E}[\tilde{T}_t \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) \tilde{T}_t] = \sum_{k=1}^K \theta_k^* \mathbb{E}[\mathbb{I}(S_t = k) \tilde{T}_t^2] = \mathbb{E}[\tilde{T}_t^2] \sum_{k=1}^K \omega_k \theta_k^*$, where $\omega_k = \frac{\mathbb{E}[\mathbb{I}(S_t = k) \tilde{T}_t^2]}{\mathbb{E}[\tilde{T}_t^2]}$ with $\sum_k \omega_k = 1$. (iii) Since $\mathbb{E}[\tilde{T}_t \mid X_t] = 0$ and $\mathbb{E}[\tilde{T}_t \mathbb{I}(S_t = k) \mid X_t] = \Delta m_k(X_t) \pi_k(X_t)$, the mean adjustment yields $\sum_{k=1}^K \theta_k^* \mathbb{E}[\bar{m}(X_t) \Delta m_k(X_t) \pi_k(X_t)]$. (iv) The confounding component evaluates to $\mathbb{E}[\tilde{T}_t (g(X_t, S_t) - \bar{g}(X_t))] = \sum_{k=1}^K \mathbb{E}[\pi_k(X_t) \Delta m_k(X_t) \Delta g_k(X_t)]$. Dividing by $\mathbb{E}[\tilde{T}_t^2]$ yields (11) and (12). ■

B Proof of Theorem 2: Semiparametric Efficiency Bound & Asymptotics

B.1 Semiparametric Efficiency Bound

Consider the semiparametric model $\mathcal{P} = \{P_\theta : \theta_k \in \Theta, \eta_k \in \mathcal{T}_\eta\}$ where the outcome equation in regime k is $Y = \theta_k T + g_k^*(X) + U$ with $\mathbb{E}[U \mid X, T, S = k] = 0$,

and treatment equation is $T = m_k^*(X) + V$ with $\mathbb{E}[V \mid X, S = k] = 0$. Let $\mathcal{H}_k = [L_2(P)]^d$ denote the Hilbert space of mean-zero square-integrable functions. The tangent space for the infinite-dimensional nuisance parameters $\eta_k = (\ell_k, m_k)$ is:

$$\mathcal{T}_{\text{nuisance}}^{(k)} = \{a(X, S) + b(X, S)V : a, b \in L_2(P), \mathbb{E}[a \mid S = k] = 0, \mathbb{E}[b \mid S = k] = 0\}. \quad (26)$$

The ordinary score for the target parameter θ_k is $S_{\theta_k}(W) = \mathbb{I}(S = k)UV$. The efficient score $S_{\theta_k}^*(W)$ is the orthogonal projection of S_{θ_k} onto the orthogonal complement of the nuisance tangent space $\left(\mathcal{T}_{\text{nuisance}}^{(k)}\right)^\perp$. Since $\mathbb{E}[S_{\theta_k} \mid X, S = k] = 0$ and $\mathbb{E}[S_{\theta_k} V \mid X, S = k] = V\mathbb{E}[UV \mid X, S = k] = 0$ by Assumption 1, S_{θ_k} is already orthogonal to $\mathcal{T}_{\text{nuisance}}^{(k)}$. Hence, the efficient score is:

$$S_{\theta_k}^*(W) = \mathbb{I}(S = k)(Y - \ell_k^*(X) - \theta_k^*(T - m_k^*(X)))(T - m_k^*(X)). \quad (27)$$

The efficient influence function is $\tilde{\psi}_k(W) = J_k^{-1} S_{\theta_k}^*(W)$ with $J_k = \mathbb{E}[\mathbb{I}(S = k)(T - m_k^*(X))^2]$. By the semiparametric convolution theorem [Bickel et al., 1993], the asymptotic variance $\sigma_k^2 = J_k^{-2} \Omega_k$ attains the semiparametric efficiency bound.

B.2 Asymptotic Normality under α -Mixing

The empirical score decomposition is $\sqrt{N}(\hat{\theta}_k - \theta_k^*) = \hat{J}_k^{-1} \frac{1}{\sqrt{N}} \sum_{t=1}^N \psi_k(W_t; \theta_k^*, \hat{\eta}_k^{(-b(t))})$. Expanding around η_k^* :

$$\frac{1}{\sqrt{N}} \sum_{t=1}^N \psi_k(W_t; \theta_k^*, \hat{\eta}_k^{(-b(t))}) = \frac{1}{\sqrt{N}} \sum_{t=1}^N \psi_k(W_t; \theta_k^*, \eta_k^*) + \frac{1}{\sqrt{N}} \sum_{t=1}^N \gamma_{tk}(\hat{\ell}_k - \ell_k^*)(\hat{m}_k - m_k^*) + \mathcal{R}_N.$$

Under Assumption 4, $\|\hat{\ell}_k - \ell_k^*\|_{L_2} \|\hat{m}_k - m_k^*\|_{L_2} = o_P(N^{-1/2})$, so the second-order remainder is $o_P(1)$. With embargo window $\tau \geq C \log N$, the covariance between $\mathcal{T}_b$ and $\mathcal{B}_b$ is bounded by $O(N\alpha(\tau)) = O(N^{1-Cc}) = o(1)$. Applying the central limit theorem for α -mixing stationary processes [Ibragimov, 1962], $\frac{1}{\sqrt{N}} \sum_{t=1}^N \psi_k(W_t; \theta_k^*, \eta_k^*) \xrightarrow{d} \mathcal{N}(0, \Omega_k)$. Slutsky's theorem completes the proof. ■

C Proof of Theorem 3: Multiway Gateaux Orthogonality to Latent Dynamics

Let $\mathbf{\Lambda} = (\mathbf{\Pi}, \pi_0, \{\mu_k, \Sigma_k\}_{k=1}^K)$ denote the finite-dimensional parameter vector of the Gaussian HMM on auxiliary state dynamics $Z_{1:N}$. The smoothed posterior probabilities are $\gamma_{tk}(\mathbf{\Lambda}) = P(S_t = k \mid Z_{1:N}; \mathbf{\Lambda})$. The expected estimating equation is:

$$\mathcal{M}_k(\theta_k, \eta_k, \mathbf{\Lambda}) \equiv \mathbb{E}[\gamma_{tk}(\mathbf{\Lambda}) \cdot ((Y_t - \ell_k^*(X_t)) - \theta_k^*(T_t - m_k^*(X_t)))(T_t - m_k^*(X_t))]. \quad (28)$$

Consider the Gateaux derivative with respect to $\mathbf{\Lambda}$ in an arbitrary direction $\mathbf{h}$:

$$\nabla_{\mathbf{\Lambda}} \mathcal{M}_k(\theta_k^*, \eta_k^*, \mathbf{\Lambda}^*)[\mathbf{h}] = \mathbb{E}[(\nabla_{\mathbf{\Lambda}} \gamma_{tk}(\mathbf{\Lambda}^*)[\mathbf{h}]) \cdot ((Y_t - \ell_k^*(X_t)) - \theta_k^*(T_t - m_k^*(X_t)))(T_t - m_k^*(X_t))]. \quad (29)$$

Applying the law of iterated expectations by conditioning on $\mathcal{F}_t = (X_t, Z_{1:N}, S_t)$:

$$\begin{aligned} \mathbb{E}[(Y_t - \ell_k^*(X_t)) - \theta_k^*(T_t - m_k^*(X_t)) \mid X_t, Z_{1:N}, S_t = k] &= \mathbb{E}[Y_t \mid X_t, S_t = k] - \ell_k^*(X_t) - \theta_k^*(\mathbb{E}[T_t \mid X_t, S_t = k]) \\ &= \ell_k^*(X_t) - \ell_k^*(X_t) - \theta_k^*(m_k^*(X_t) - m_k^*(X_t)) = 0. \end{aligned}$$

Since $\nabla_{\mathbf{\Lambda}} \gamma_{tk}(\mathbf{\Lambda}^*)[\mathbf{h}]$ is a deterministic function of $Z_{1:N}$, it pulls out of the inner expectation:

$$\nabla_{\mathbf{\Lambda}} \mathcal{M}_k(\theta_k^*, \eta_k^*, \mathbf{\Lambda}^*)[\mathbf{h}] = \mathbb{E}[(\nabla_{\mathbf{\Lambda}} \gamma_{tk}(\mathbf{\Lambda}^*)[\mathbf{h}]) \cdot \mathbf{0}] = 0. \quad (30)$$

Because the Gateaux derivative vanishes identically, $\sqrt{N}$ -estimation of $\mathbf{\Lambda}$ has zero first-order influence on the asymptotic distribution of $\hat{\theta}_k$. ■

D Proof of Theorem 4: Oracle Equivalence

Let $\hat{\theta}_k^{\text{oracle}}$ denote the estimator using true indicator $\mathbb{I}(S_t = k)$. The difference in scores is:

$$\frac{1}{\sqrt{N}} \sum_{t=1}^N (\gamma_{tk} - \mathbb{I}(S_t = k))(Y_t - \ell_k^*(X_t) - \theta_k^*(T_t - m_k^*(X_t)))(T_t - m_k^*(X_t)). \quad (31)$$

By Cauchy-Schwarz and the assumption $\frac{1}{N} \sum_{t=1}^N \|\gamma_t - \mathbb{I}_{S_t}\|_1 = o_P(N^{-1/4})$:

$$\left| \frac{1}{\sqrt{N}} \sum_{t=1}^N (\gamma_{tk} - \mathbb{I}(S_t = k)) \tilde{Y}_{tk} \tilde{T}_{tk} \right| \leq \sqrt{N} \cdot \left(\frac{1}{N} \sum_{t=1}^N (\gamma_{tk} - \mathbb{I}(S_t = k))^2 \right)^{1/2} \cdot o_P(1) = \sqrt{N} \cdot o_P(N^{-1/2}) = o_P(1) \quad (32)$$

Thus, $\sqrt{N}(\hat{\theta}_k - \hat{\theta}_k^{\text{oracle}}) = o_P(1)$. ■

E Proof of Proposition 5: Markov Occupation Variance Decomposition

The sample Average Treatment Effect is $\hat{\theta}_{\text{SATE}} = \sum_{k=1}^K \bar{\gamma}_k \hat{\theta}_k = \bar{\gamma}^\top \hat{\theta}$, where $\bar{\gamma} = \frac{1}{N} \sum_{t=1}^N \gamma_t$. Conditional on the realized regime trajectory $S_{1:N}$, the conditional asymptotic variance is $\sigma_{\text{cond}}^2 = \bar{\gamma}^\top \Sigma \bar{\gamma}$, where $\Sigma = J^{-1} \Omega J^{-1} / N$. The population estimand is $\theta_{\text{PATE}}^* = \sum_{k=1}^K \pi_k^* \theta_k^* = \boldsymbol{\pi}^{*\top} \boldsymbol{\theta}^*$. By the law of total variance:

$$\begin{aligned} \text{Var}(\hat{\theta}_{\text{PATE}}) &= \mathbb{E} \left[\text{Var}(\hat{\theta}_{\text{PATE}} \mid S_{1:N}) \right] + \text{Var} \left(\mathbb{E}[\hat{\theta}_{\text{PATE}} \mid S_{1:N}] \right) \\ &= \bar{\gamma}^\top \Sigma \bar{\gamma} + \boldsymbol{\theta}^{*\top} \text{Var}(\bar{\gamma}) \boldsymbol{\theta}^*. \end{aligned} \quad (33)$$

For an irreducible, aperiodic Markov chain with transition matrix $\mathbf{\Pi}$ and stationary distribution $\boldsymbol{\pi}^*$, the fundamental matrix [Kemeny and Snell, 1976] is $\mathbf{Z} = (\mathbf{I} - \mathbf{\Pi} + \mathbf{1}\boldsymbol{\pi}^{*\top})^{-1}$. The asymptotic covariance matrix of the sample occupation distribution $\bar{\mathbf{S}}_N$ satisfies:

$$\lim_{N \rightarrow \infty} NCov(\bar{\mathbf{S}}_N) = \mathbf{D}_\pi \mathbf{Z} + \mathbf{Z}^\top \mathbf{D}_\pi - \mathbf{D}_\pi - \boldsymbol{\pi}^* \boldsymbol{\pi}^{*\top}, \quad (34)$$

where $\mathbf{D}_\pi = \text{diag}(\boldsymbol{\pi}^*)$. For a two-state Markov chain ($K = 2$) with persistence parameter $\rho = 1 - P_{01} - P_{10} = \lambda_2(\mathbf{\Pi})$:

$$\text{Var}(\bar{S}_N) = \frac{\pi_0^* \pi_1^*}{N} \left(\frac{1 + \rho}{1 - \rho} \right) + o(N^{-1}). \quad (35)$$

Since $\theta_{\text{ATE}}(S) = \theta_0^* + (\theta_1^* - \theta_0^*)\bar{S}_N$, multiplying by $(\Delta\theta^*)^2 = (\theta_1^* - \theta_0^*)^2$ yields the total population variance $\sigma_{\text{total}}^2 = \sigma_{\text{cond}}^2 + \text{Var}(\bar{S}_N)(\Delta\theta^*)^2$. ■

F WHO 2021 Concentration-Response & Policy Valuation

Health impact calculations follow the official WHO 2021 Global Air Quality Guidelines systematic meta-analysis for all-cause non-accidental mortality. The attributable annual deaths avoided ΔM from an annual mean $\text{PM}_{2.5}$ reduction $\Delta \text{PM}_{2.5}$ is:

$$\Delta M = y_0 \cdot \text{Pop} \cdot (1 - \exp(-\beta \cdot \Delta \text{PM}_{2.5})), \quad (36)$$

where baseline mortality $y_0 = 7.2/1000$ per year (Registrar General of India SRS), and $\beta = \ln(1.062)/10.0 \approx 0.006015$ corresponds to pooled relative risk $\text{RR} = 1.062$ per $10 \mu\text{g}/\text{m}^3$. Economic evaluation applies a Value of Statistical Life (VSL) of \$0.45M USD per life saved, grounded in Indian empirical environmental health economics [Dominici et al., 2014].